

# Causal Inference

Xin Zhang

Peking University

Most of the content is from Chapter 1 of “Causality”  
second edition by Judea Pearl and  
“Actual Causality: A Survey” by Joseph Halpern

# Recap of Last Lecture

- Probabilistic Logic Programming
  - Logic programming + probabilities
  - Unifying logic and probabilities
    - Logic: Expressiveness
    - Probabilities: Handling uncertainty

# Recap of Last Lecture

- Representative language: Problog
  - $\text{Problog} = \text{Datalog} + \text{Probabilities} + \text{Additional features}$

# Problog: Example Program

0.5 :: stayUp.

0.7 :: drinkCoffee :- stayUp.

0.5 :: drinkCoffee :- \+ stayUp.

0.9 :: fallSleep :- \+ drinkCoffee, stayUp.

0.3 :: fallSleep :- drinkCoffee, stayUp.

0.1 :: fallSleep :- \+ stayUp.

evidence(fallSleep).

query(stayUp).

# Problog: Semantics

- First, ground the program into a Boolean program
- The Boolean program describes a distribution of Datalog program, which in turn defines a distribution of outputs

# Semantics of Problog

- Ground

Constants: 0, 1, 2, 3 4

$\text{path}(A,C) \text{ :- path}(A,B), \text{edge}(B,C), \text{r}(A,B,C).$

Generates

$\text{path}(0,0) \text{ :- path}(0,0), \text{edge}(0,0), \text{r}(0,0,0).$

$A=0, B=0, C=0$

$\text{path}(0,1) \text{ :- path}(0,0), \text{edge}(0,1), \text{r}(0,0,1).$

$A=0, B=0, C=1$

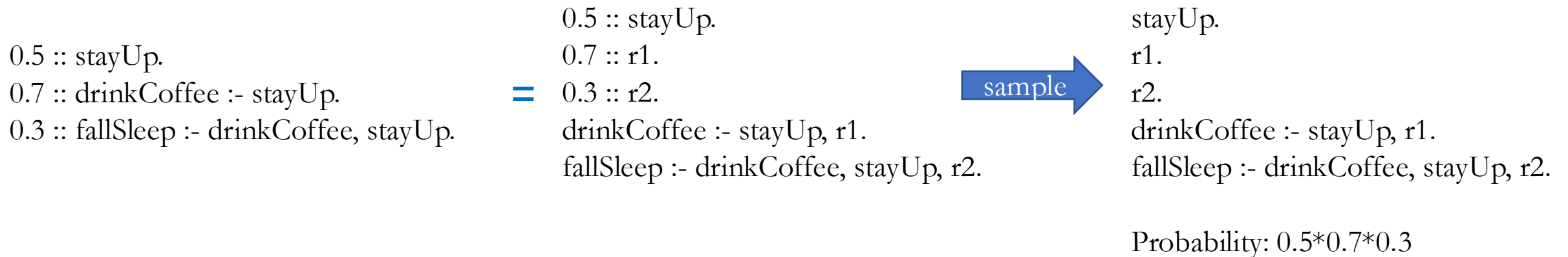
$\text{path}(0,1) \text{ :- path}(0,0), \text{edge}(0,1), \text{r}(0,0,1).$

$A=0, B=0, C=1$

...

# Semantics of Problog

- From a Problog program, we can sample a Datalog program by sampling the facts



# Solving

- Once we have a grounded program, we can leverage existing techniques
- Idea 1: convert the program into a Bayesian network
- Idea 2: convert the program into a Boolean formula with weights (MaxSAT)



# Solving: Converting into a MaxSAT

- Finding the most likely solution becomes solving the MaxSAT
- Computing marginal probabilities becomes weighted model counting

# This Class

- Causal inference
  - Structural equation model (Pearl)
  - Causal inference in probabilistic programming
  - Actual causality
- Not causal discovery
  - Assume we have a model
  - How to use the model to represent causality
  - How to reason with the model

# Motivating Example

- If a person has long hair, they are likely to be a girl
- If we change a boy from short hair to long air, would he become a girl?

## **Intervention**

# Question

- Can we separate causality from correlation without intervention?

# Motivating Example

- Xiaoming was late for the lecture. Would he still be late for the lecture if he had got up at 6am?

**Counterfactual**

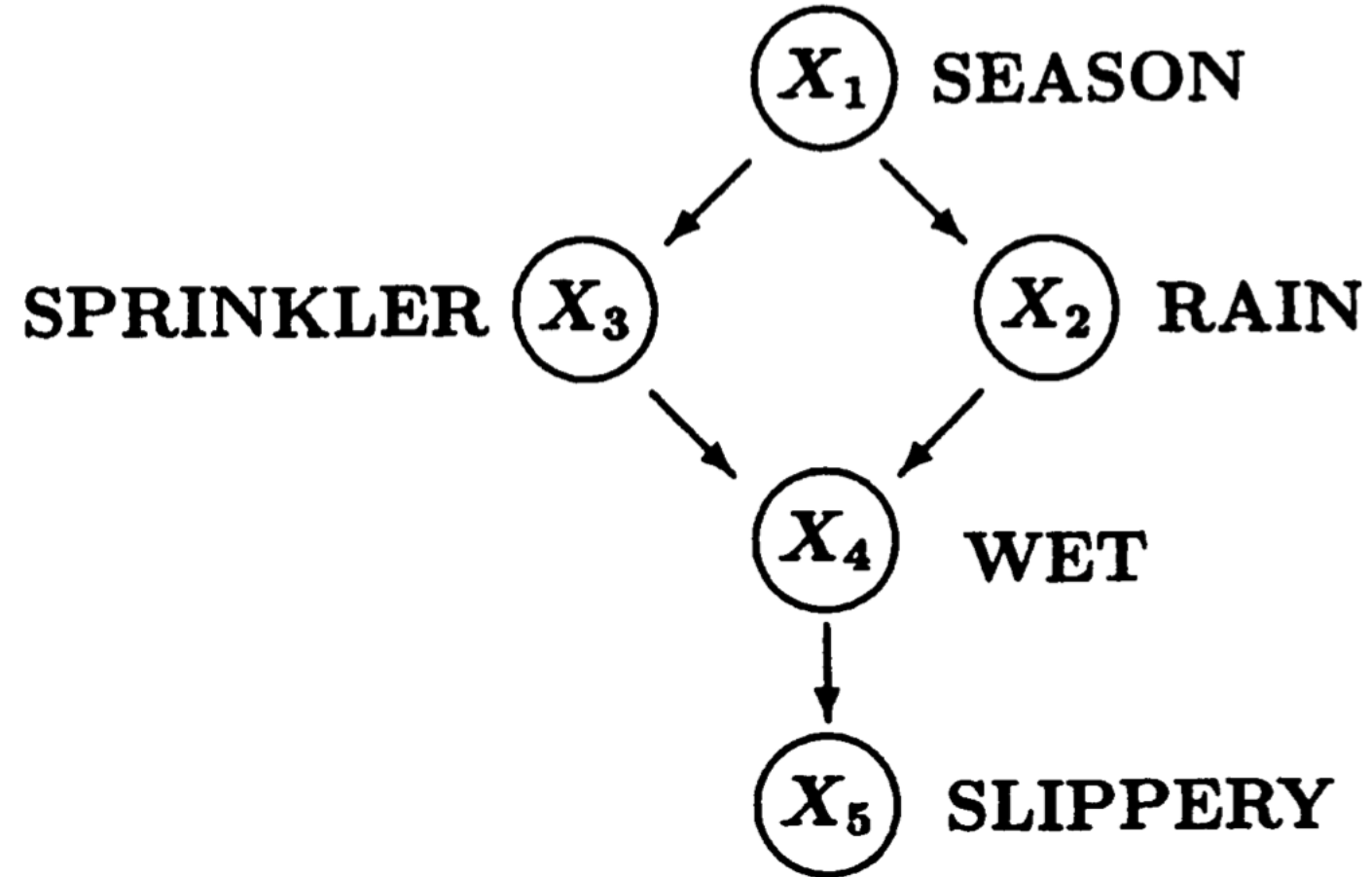
# Pearl's Causal Hierarchy

- L1: Predictions: What if I observe ... ?
- L2: Interventions: What if I change ... ?
- L3: Counterfactuals: What if we did ... given ... ?

What models can  
be used to answer  
these questions?

# Causal Bayesian Network: Handling Interventions

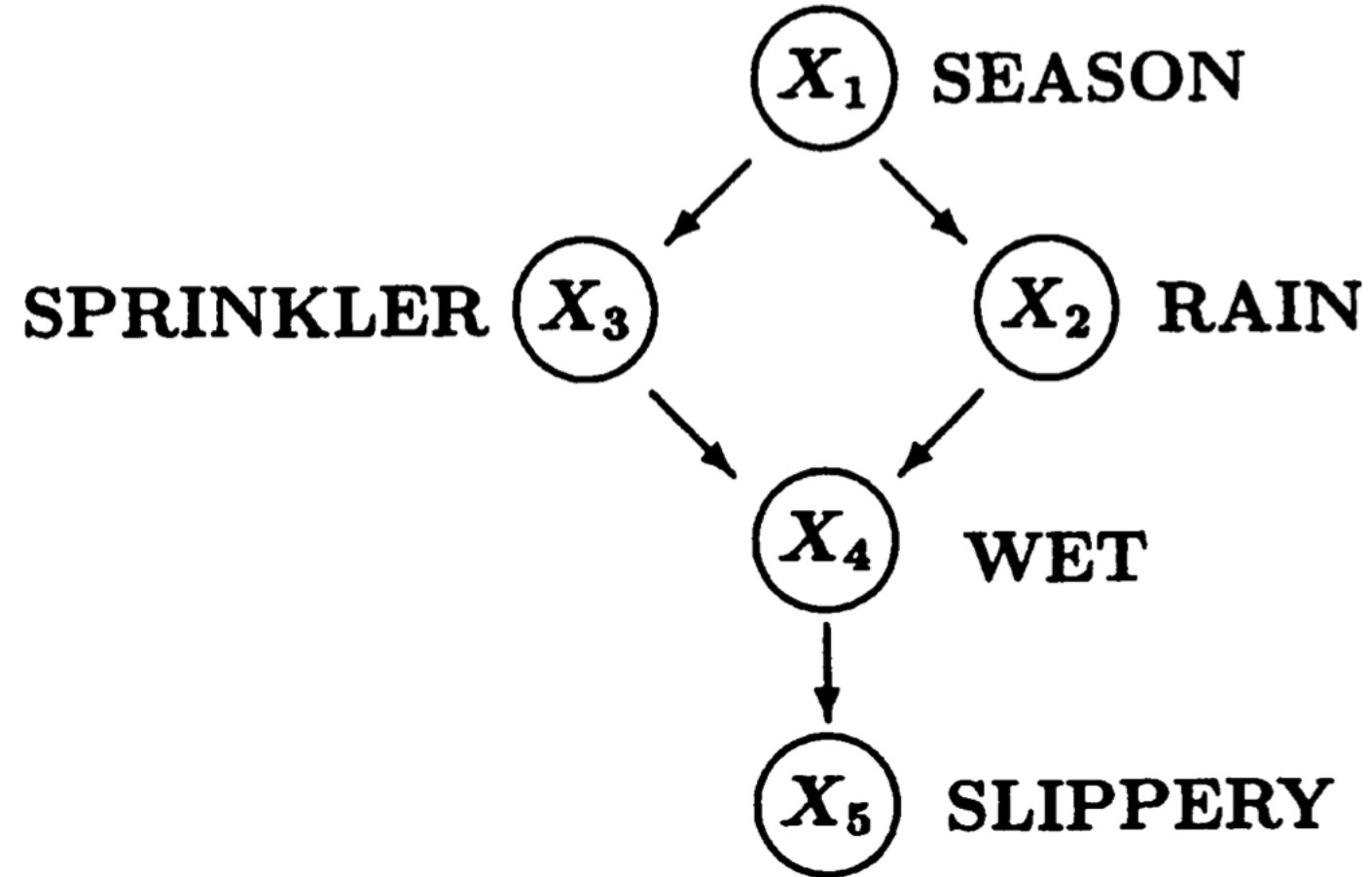
What is the joined probability distribution if we observe the sprinkler is on?



# Causal Bayesian Network: Handling Interventions

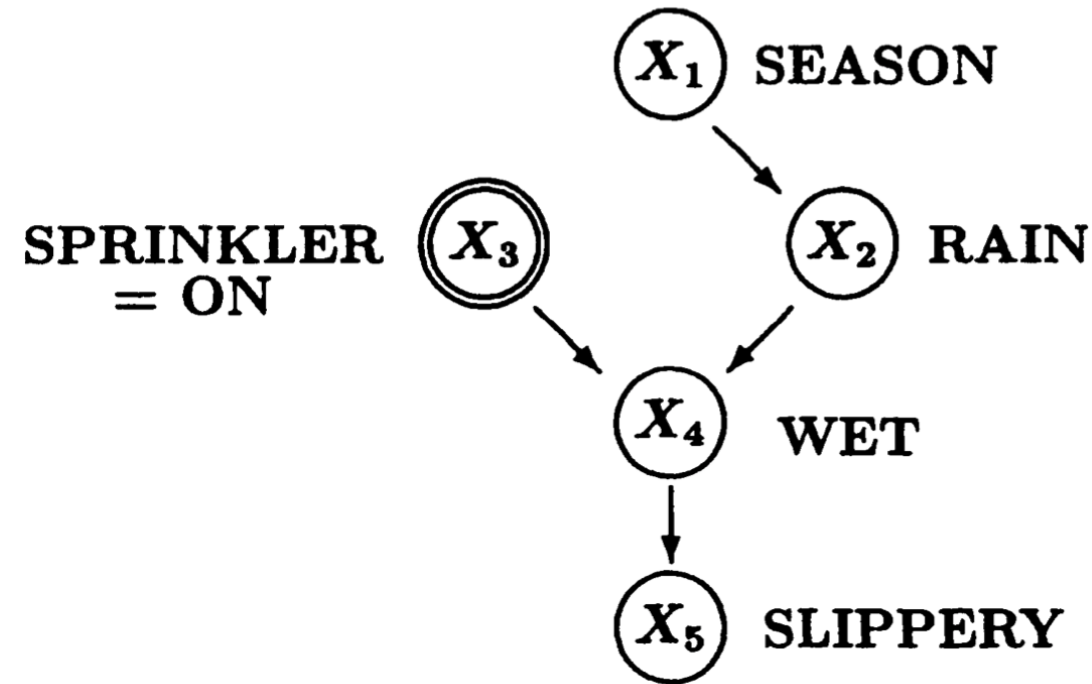
What is the joint probability  
if we intervene on the  
sprinkler by turning it on?

$\text{do}(X_3=\text{On})$





# Causal Bayesian Network: Handling Interventions



$$P_{X_3=\text{On}}(x_1, x_2, x_4, x_5) = P(x_1) P(x_2 | x_1) P(x_4 | x_2, X_3 = \text{On}) P(x_5 | x_4),$$

# Causal Bayesian Network: Handling Interventions

## Definition 1.3.1 (Causal Bayesian Network)

Let  $P(v)$  be a probability distribution on a set  $V$  of variables, and let  $P_x(v)$  denote the distribution resulting from the intervention  $do(X = x)$  that sets a subset  $X$  of variables to constants  $x$ . Denote by  $P_*$  the set of all interventional distributions  $P_x(v)$ ,  $X \subseteq V$ , including  $P(v)$ , which represents no intervention (i.e.,  $X = \emptyset$ ). A DAG  $G$  is said to be a **causal Bayesian network** compatible with  $P_*$  if and only if the following three conditions hold for every  $P_x \in P_*$ :

# Causal Bayesian Network: Handling Interventions

(i)  $P_x(v)$  is Markov relative to  $G$ ;

Conditional Independence

(ii)  $P_x(v_i) = 1$  for all  $V_i \in X$  whenever  $v_i$  is consistent with  $X = x$ ;

(iii)  $P_x(v_i|pa_i) = P(v_i|pa_i)$  for all  $V_i \notin X$  whenever  $pa_i$  is consistent with  $X = x$ .

# Defining Effects of Interventions

The distribution  $P_x(v)$  resulting from the intervention  $do(X = x)$  is given as a **truncated-factorization**

$$P_x(v) = \prod_{\{i|V_i \notin X\}} P(v_i|pa_i) \text{ for all } v \text{ consistent with } x, \quad (1.37)$$

# Defining Effects of Interventions

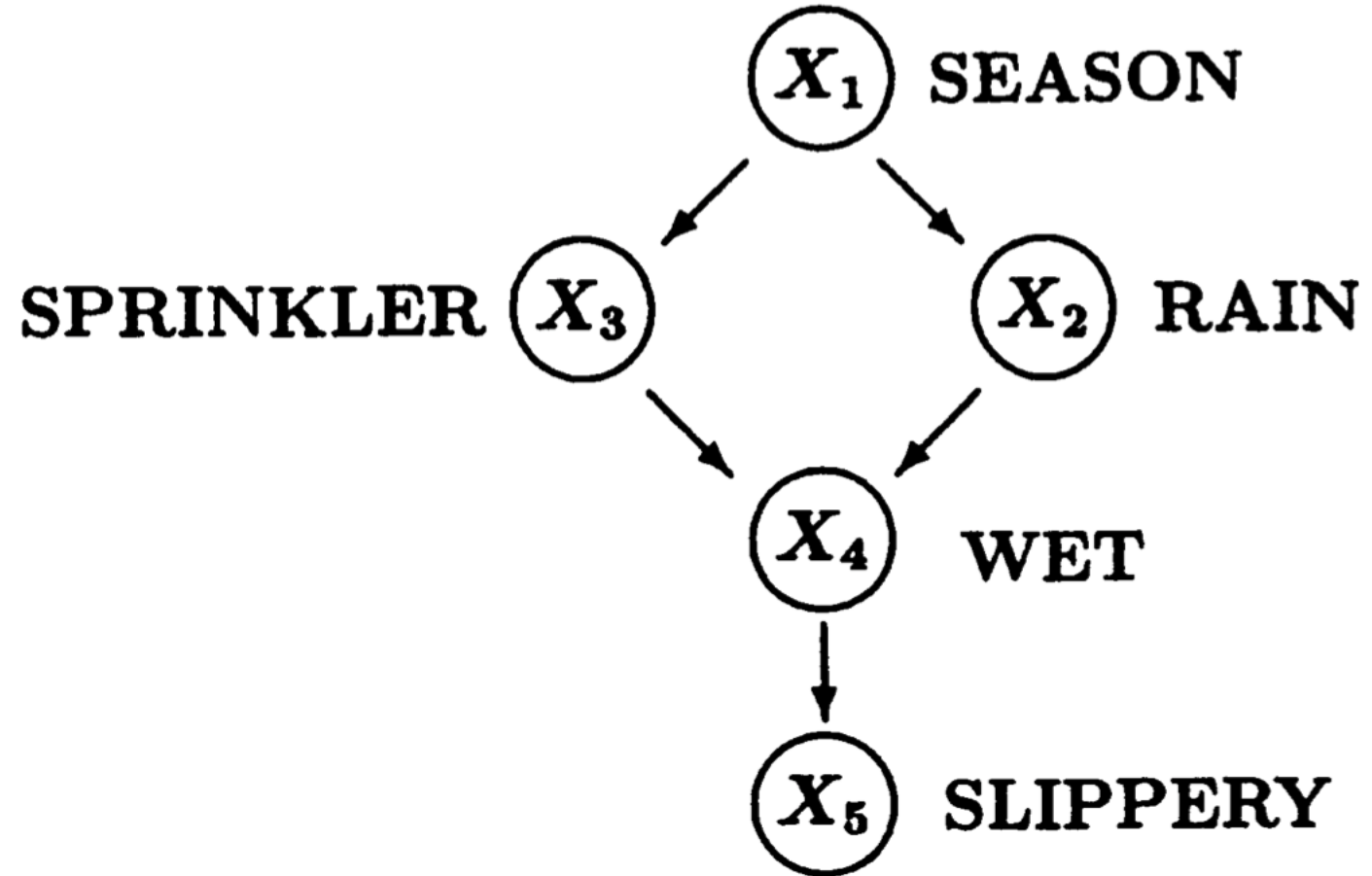
- On the graph:
  - Cut the connections from the parents to the intervened nodes
  - Set the intervened nodes to the corresponding values

# Advantages of Using a Graphical Model

- Modular
- Can use tools like d-separation to reason about the impact of interventions

# What About Counterfactuals?

Given the grass is slippery, will it still be slippery if we had turned off the sprinkler?



# Structural Equation (Functional) Model

- Functional causal model
  - Can answer all three questions
- Expressed using deterministic functional equations
  - Probabilities are introduced by assuming certain variables are unobserved
  - Follows Laplace's conception of natural phenomena
- Advantages over stochastic representations
  - More general
  - More in tune with human intuition
  - Counterfactuals



# Structural Equations

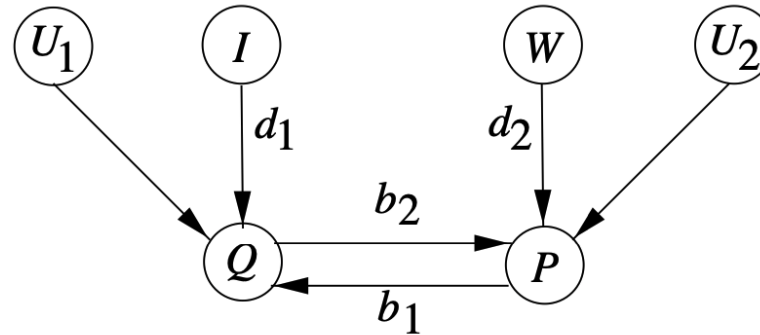
- A functional causal model consists a set of equations:

$$x_i = f_i(\underline{pa_i}, \underline{u_i}), \quad i = 1, \dots, n,$$

parents

Errors due to  
omitted factors.  
Random.

# Structural Equations: Example I



**Figure 1.5:** Causal diagram illustrating the relationship between price ( $P$ ), demand ( $Q$ ), income ( $Z$ ), and wages ( $W$ ).

$$q = b_1 p + d_1 i + u_1, \quad (1.42)$$

$$p = b_2 q + d_2 w + u_2, \quad (1.43)$$

# Structural Equations: Example II

Explicitly separate deterministic parts from the stochastic parts

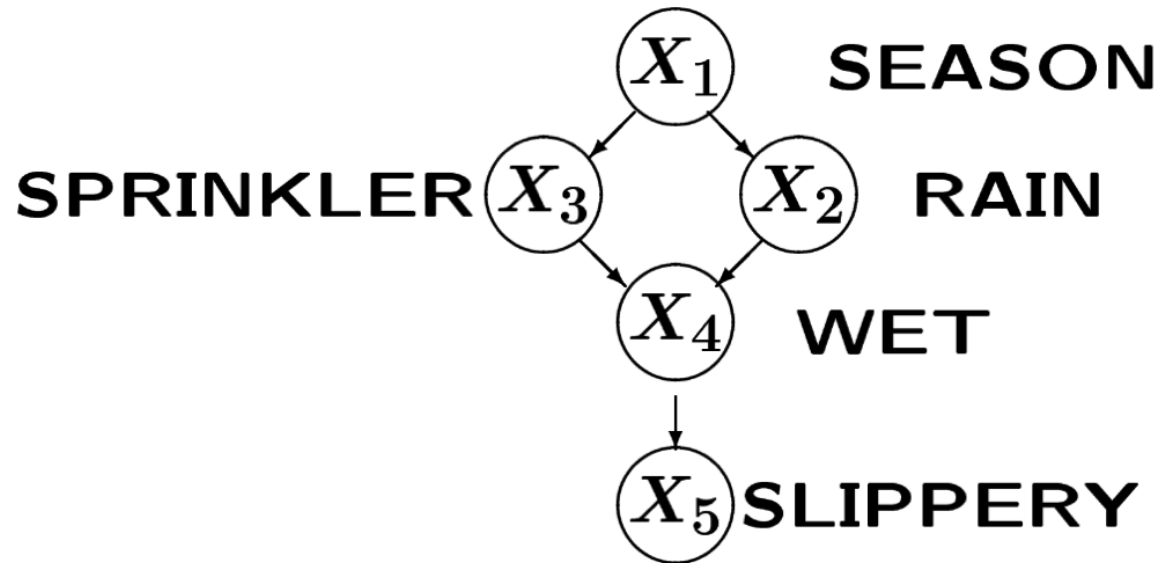


Figure 1.2

$$\begin{aligned} x_1 &= u_1, \\ x_2 &= f_2(x_1, u_2), \\ x_3 &= f_3(x_1, u_3), \\ x_4 &= f_4(x_3, x_2, u_4), \\ x_5 &= f_5(x_4, u_5). \end{aligned}$$

$$\begin{aligned} x_2 &= [(X_1 = \text{winter}) \vee (X_1 = \text{fall}) \vee u_2] \wedge \neg u'_2, \\ x_3 &= [(X_1 = \text{summer}) \vee (X_1 = \text{spring}) \vee u_3] \wedge \neg u'_3, \\ x_4 &= (x_2 \vee x_3 \vee u_4) \wedge \neg u'_4, \\ x_5 &= (x_4 \vee u_5) \wedge \neg u'_5, \end{aligned} \tag{1.45}$$

# Goal: Handle the Whole Pearl's Causal Hierarchy

- L1: Predictions: What if I observe ... ?

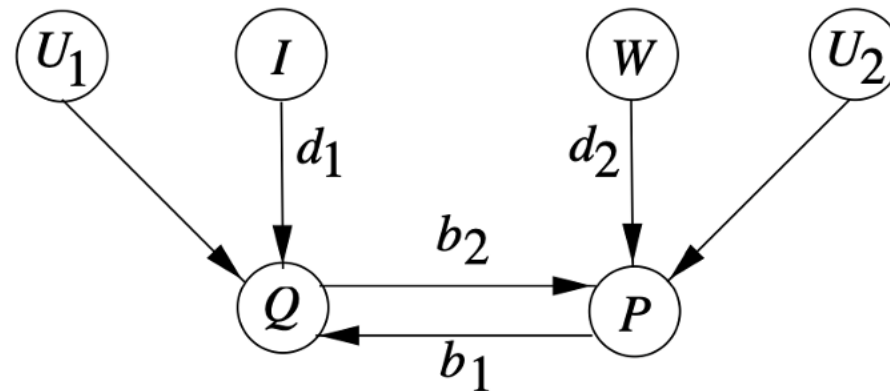
- L2: Interventions: What if I change ... ?

What models can  
be used to answer  
these questions?

- L3: Counterfactuals: What if we did ... given ... ?

# Probabilistic Predictions in Causal Models

- Causal diagram:



- Semi-Markovian model: the diagram is acyclic
- Markovian model: the diagram is acyclic and the errors are independent

# The Causal Markov Condition

## **Theorem 1.4.1 (Causal Markov Condition)**

Every Markovian causal model  $M$  induces a distribution  $P(x_1, \dots, x_n)$  that satisfies the parental Markov condition relative the causal diagram  $G$  associated with  $M$ ; that is, each variable  $X_i$  is independent on all its non-descendants, given its parents  $PA_i$  in  $G$  (Pearl and Verma 1991)

# The Causal Markov Condition Follows two Causal Assumptions

- Include every variable that is the cause of two or more variables in the model (not in the background)
- Reichenbach's common-cause assumption
  - No correlation without causation
  - If any two variables are dependent, then one is the cause of the other or there is a third variable causing both (confounder)

# Interventions and Causal Effects in Functional Models

- Simply modify the corresponding equations

$$x_3 = f(x_1, u_3) \rightarrow x_3 = 0n$$

- More formally: fix the intervened variables to their specified values, and removing equations defining them
- Intervening on a causal Markovian model is the same as intervening on a causal Bayesian network



# Advantages Over Causal Bayesian Networks

- Extensions to feedback systems and non-Markovian models
- Modifications of parameters are meaningful
  - Functions generate the joint distribution, conditional probabilities are then inferred
- Simplifying the analysis of causal effects
- Permit the analysis of context-specific actions and policies

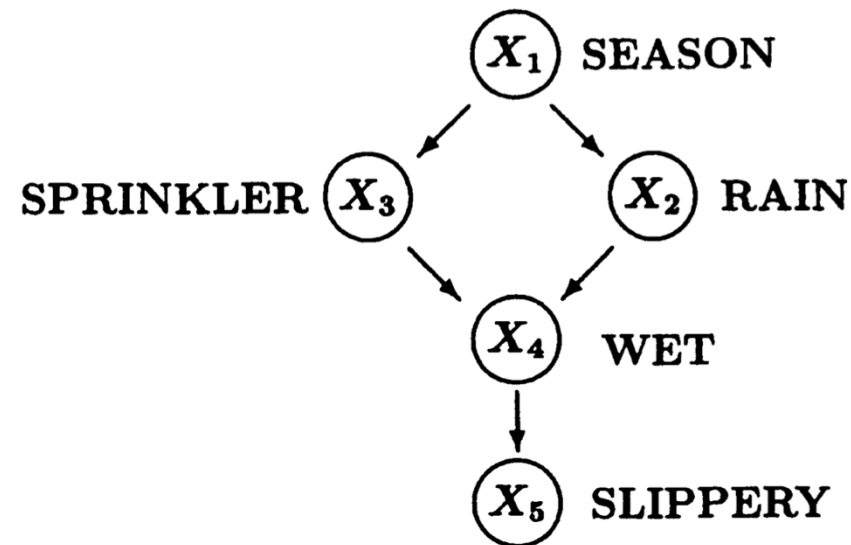
# Last Point Explained

- Interventions affect contexts
  - Example: the patient has been examined by the doctor and he has some symptoms, but now the new intervention will affect these symptoms
- We will see that counterfactuals are similar

# Counterfactuals in Functional Models

- Causal Bayesian networks have trouble dealing with counterfactuals
  - The simplest example:
    - Consider two independent boolean variables  $x$  and  $y$ , we have  $P(x|y) = 0.5$ , given  $y = 1$ , what is  $P(y = 1 \mid \text{do}(x) = 0, y = 1)$ ?
  - A more complex example:

$\text{do}(x_3 = \text{ON})$ ,  $X_5 = \text{True}$



# Understand Counterfactuals Better

- Counterfactuals can be seen as the combination of conditioning and interventions:
  - Use observations to infer the posterior distributions of the hidden variables
  - Based on the posterior distributions, predict under interventions

# Three Steps for Computing

For computing  $P(Y=y \mid \text{do}(X=x), e)$ :

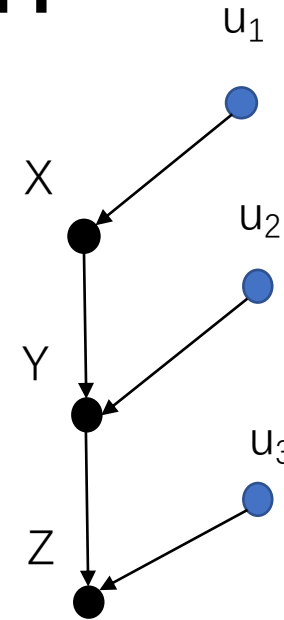
1. (abduction): Update the probability  $P(u)$  to obtain  $P(u \mid e)$
2. (action): Perform intervention  $\text{do}(X) = x$
3. (prediction) Use the modified model to compute  $P(Y=y)$

# More on Computing Counterfactuals

- A major difficulty of the previous approach is the need to compute and store  $p(u | e)$
- Can we overcome this problem by leveraging algorithms in graphical models?

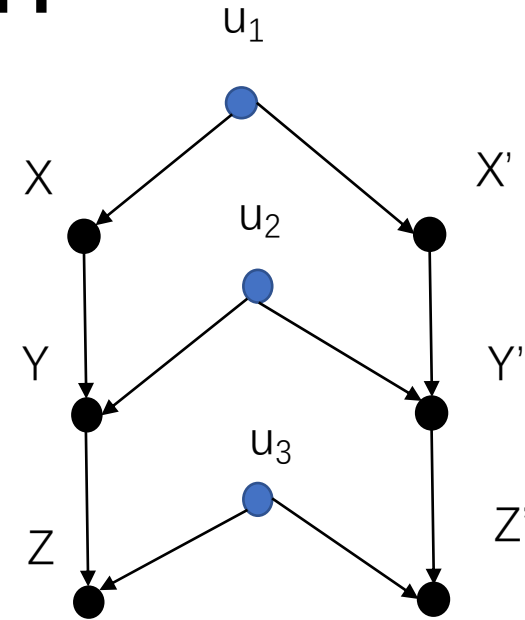
# The Twin Network Approach

- Consider the following example
  - $X = u_1, Y = X + u_2, Z = Y + u_3$
- How to compute  $P(Z \mid \text{do}(X) = x, Z=z)$ ?



# The Twin Network Approach

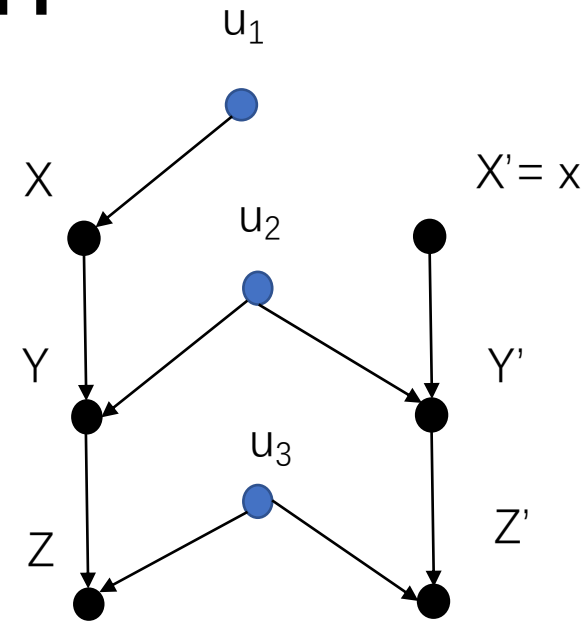
- $P(Z \mid \text{do}(X) = x, Z=z)$  becomes  $P(Z' \mid \text{do}(X') = x, Z'=z)$





# The Twin Network Approach

- $P(Z \mid \text{do}(X) = x, Z=z)$  becomes  $P(Z' \mid X' = x, Z'=z)$



# The Twin Network Approach

- Duplicate all the equations and observed variables
- Perform intervention on the copied part
- Keep observations on the original part

Can you apply the twin network approach to causal Bayesian networks?

# Two Mainstream Causal Models

- Structural equation model (Pearl)
  - This class
- Potential outcomes (Neyman-Rubin)
- Two models are theoretically equivalent, but have their own advantages in practice

# Causal inference in probabilistic programming

- A Language for Counterfactual Generative Models. Zenna Tavares, James Koppel, Xin Zhang, Ria Das, Armando Solar-Lezama. ICML 2021
- Implicitly implements the twin network approach
  - Lazy evaluation
  - Stores the program piece that computes a given variable

# Actual Causality

- Interventions and counterfactuals basically tells how a things changes in response how another thing changes
- But it doesn't define what is the cause/reason of something.
- Causality answers this

# Some Heads-Up

- Two notions of causality
  - Type (general) causality: smoking causes lung cancer
  - Actual causality: the fact that David smoked like a chimney for 30 years caused him to get cancer last year
- Actual causality is a long-debated problem in philosophy, math, and computer science
- We are not going to include philosophical discussions
  - No chicken-or-egg problems
- We assume there is a known model of the world and discuss how to define actual causalities according to it
  - Causes can be different if the modeling is different

# The Big Picture on Actual Causalities

- The definition has changed many times
- No satisfying answers
- The new definitions are usually invented in response to counterexample

# The Big Picture on Actual Causalities

- Attempts to define causality goes back to Aristotle
- Relatively recent trend (Lewis 1973) is to use counterfactuals
- More recent: capture counterfactuals using structural equations
- Pearl & Halpern definitions:
  - UAI 2001
  - BJPS 2005



# But-For Causes

- Jimmy threw a ball to shatter the bottle
  - $\text{JimmyThrows} = u_1$
  - $\text{BottleShatters} = \text{JimmyThrows}$
- If Jimmy doesn't throw the ball, the bottle won't shatter
  - Therefore Jimmy throwing the ball is the cause for the bottle to shatter

# But-For Causes

- Counter-example (preemption): Suzy and Jimmy both pick up rocks and throw them at a bottle. Suzy's rock gets there first, shattering the bottle. Since both throws are perfectly accurate, Jimmy's would have shattered the bottle if Suzy's throw had not preempted it
- JimmyThrows = u1, SuzyThrows = u2,  
SuzyShatters = SuzyThrows,  
JimmyShatters = JimmyThrows & !SuzyShatters,  
BottleShatters = SuzyShatters | JimmyShatters

# Pearl and Halpern's: Problem Setting

- Represent the model using structural equations
- Remove all randomness by fixing the unobserved variables
  - In other words, the causes are defined for specific contexts
- The cause can be any conjunction of primitive events
- Arbitrary Boolean combinations of primitive events can be caused

# Pearl and Halpern's Definition

- $\vec{X} = \vec{x}$  is an actual cause of  $\phi$  in situation  $(M, \vec{u})$  if
  - $AC1. (M, \vec{u}) \models (\vec{X} = \vec{x}) \wedge \phi$ 
    - Both  $(\vec{X} = \vec{x})$  and  $\phi$  are true in the actual world
  - $AC2.$  Complicated. Captures counterfactuals
  - $AC3.$   $\vec{X}$  is minimal; no subset of  $\vec{X}$  satisfies  $AC1$  and  $AC2.$ 
    - No irrelevant conjuncts

# Pearl and Halpern's Definition

- AC2. There is a set of  $\vec{W}$  of variables in  $V$  and a setting  $\vec{x}'$  of the variables in  $\vec{X}$  such that if  $(M, \vec{u}) \models (\vec{W} = \vec{w})$ , then
 
$$(M, \vec{u}) \models \left( \vec{X} \leftarrow \vec{x}', \vec{W} \rightarrow \vec{w} \right) \wedge \neg \phi$$

In words: keeping the variables in  $\vec{W}$  fixed at their actual values, changing  $\vec{X}$  can change the outcome  $\phi$

# Example

- JimmyThrows = u1, SuzyThrows = u2,  
 SuzyShatters = SuzyThrows,  
 JimmyShatters = JimmyThrows & !SuzyShatters,  
 BottleShatters = SuzyShatters | JimmyShatters

Let  $\vec{X} = \{SuzyThrows\}$ ,  $\vec{W} = \{JimmyShatters\}$ ,  $\phi = BottleShatters$ ,  
 then  $(M, \vec{u}) \models (\vec{X} \leftarrow \vec{x}, \vec{W} \rightarrow \vec{w}) \wedge \neg\phi$

# Another Example

- Suppose in an election, Jim will be elected if two of the three voters vote for him.
- None of the voters voted for Jim. What is a cause of Jim not being elected?
- For more, watch <https://www.youtube.com/watch?v=hXnCX2pJ0sg>